

Seeing the Invisible

Physics-Guided Missing-Physics Learning for PEMFC Flooding Diagnosis

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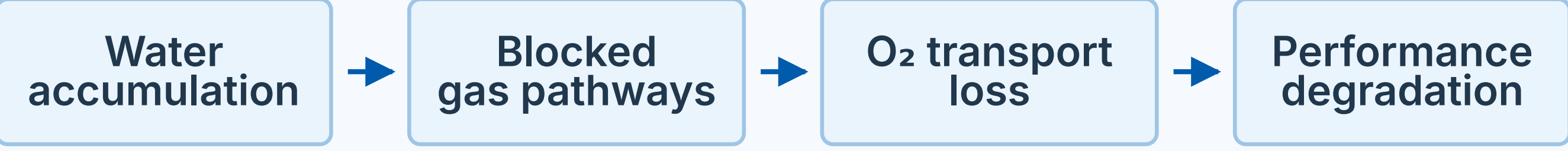


ABSTRACT

Liquid water accumulation in porous gas diffusion layers critically limits mass transport in polymer electrolyte membrane fuel cells, yet the internal flooding state is difficult to observe directly. More broadly, physics-based multiphysics models often rely on empirical or fitted constitutive relations to represent unresolved transport behavior, making constitutive uncertainty a major source of prediction error. Here, we propose a physics-informed grey-box framework that retains the known governing equations within a finite-element solver while learning only the missing constitutive physics. A learnable gas-phase transport closure is embedded directly into the two-phase FEM model and identified from synthetic reference responses using discrete-adjoint sensitivities, without prescribing a fixed functional form in advance. After training at RH 50%, the learned constitutive law is frozen and evaluated at unseen RH 70% on two discretizations, 16×40 and 32×80 . Agreement with the synthetic reference is retained without retraining in the reported test cases, supporting transfer of the learned closure across the tested humidity and discretization changes. These synthetic tests motivate future experimental validation for flooding diagnosis and physics-guided digital twins.

INTRODUCTION

Why flooding matters



Challenges in flooding diagnosis

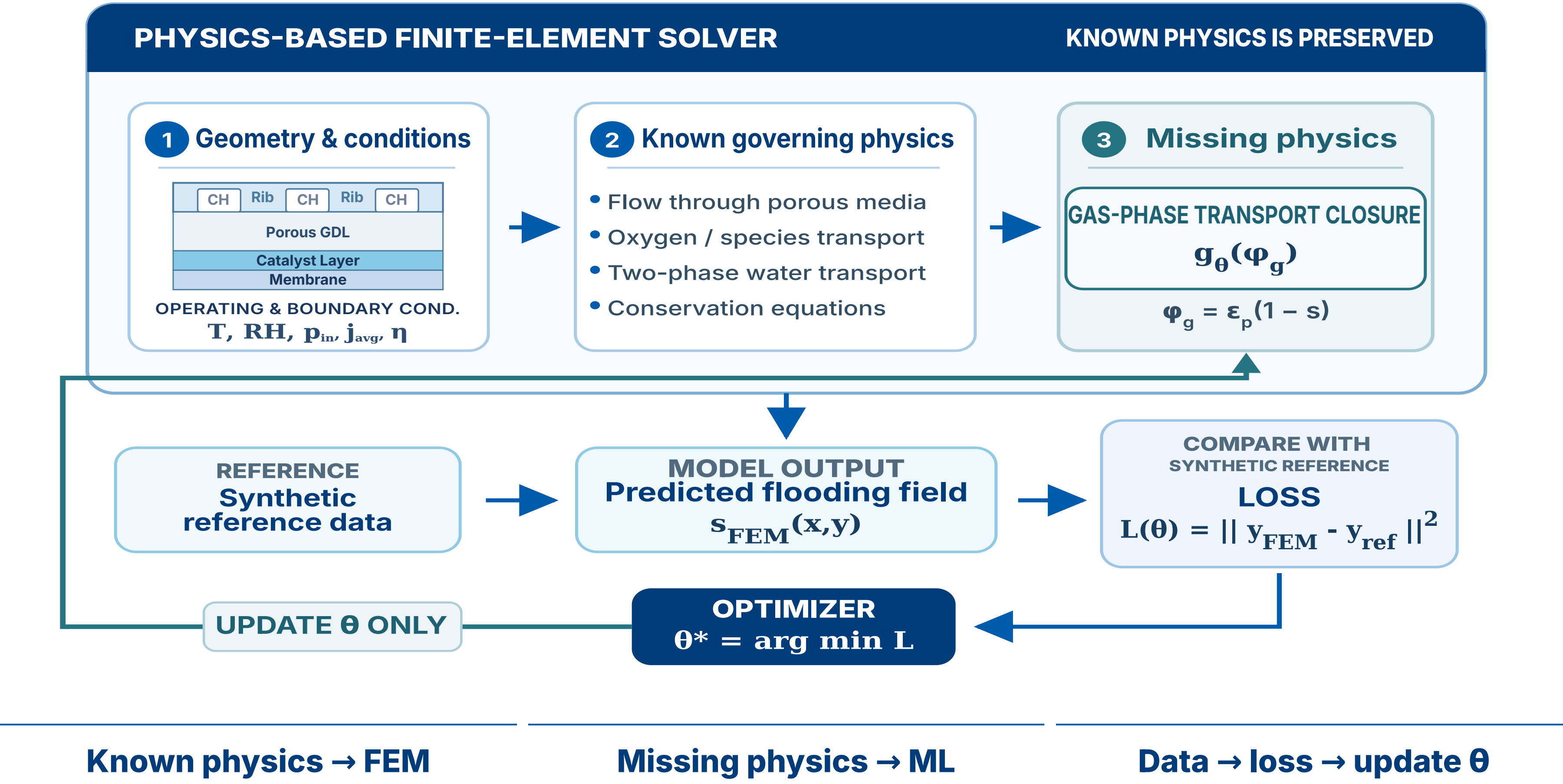
Observation	Internal liquid-water saturation $s(x,y)$ cannot be directly and continuously observed under realistic operating conditions.
Modeling	Flooding predictions depend strongly on uncertain constitutive relations such as $D_{\text{eff}}(s)$, $k_r(s)$, and $p_c(s)$.

Existing ML vs our objective

DIFFERENT LEARNING TARGETS, NOT A BETTER/WORSE RANKING

PINN Learn PDE solution Physics is enforced mainly through residual-based training.	FNO / Neural Operator Learn solution operator Maps input fields or parameters to solution fields.	OUR APPROACH FEM + Missing Physics ML Learn the unknown physical relation Known conservation laws remain in FEM; only uncertain constitutive physics is learned.
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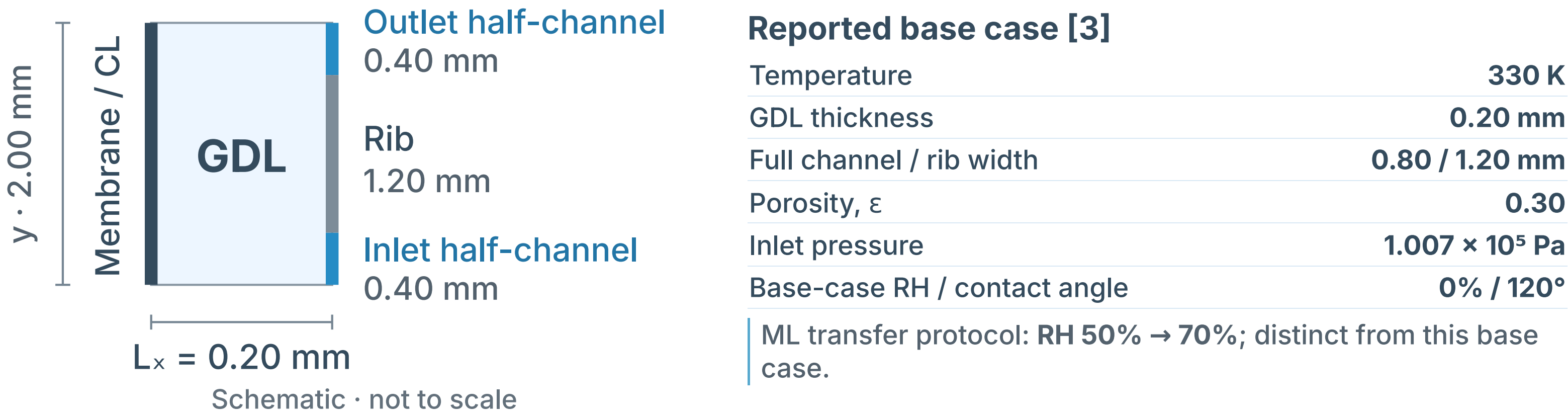
METHODS



RESULTS

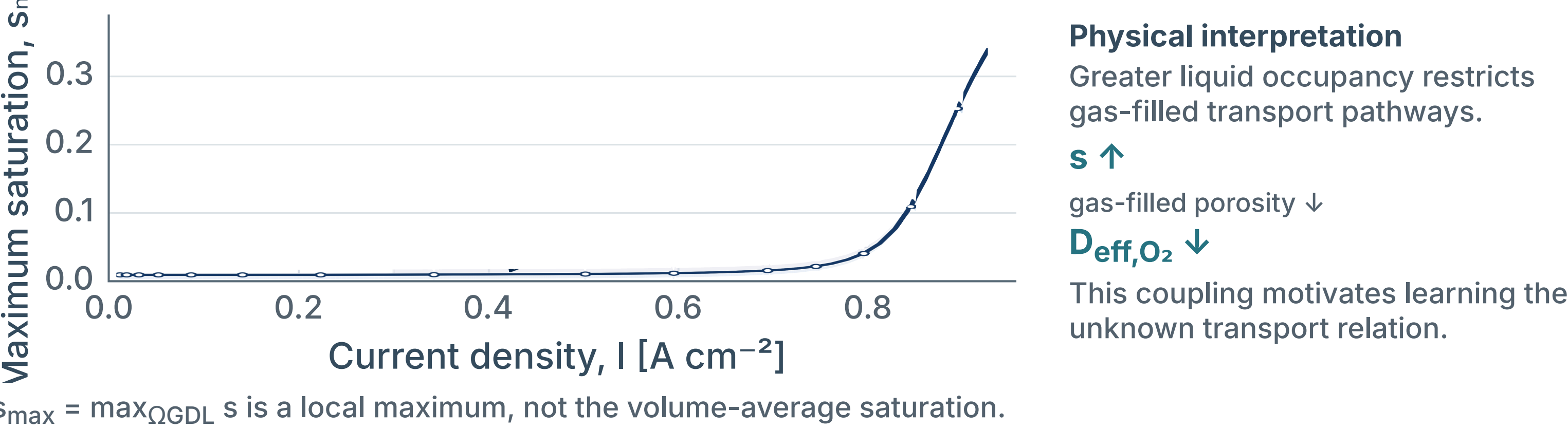
01 Model domain & reference conditions

A two-dimensional GDL unit cell resolves transport between the membrane/CL and the channel-rib boundary.



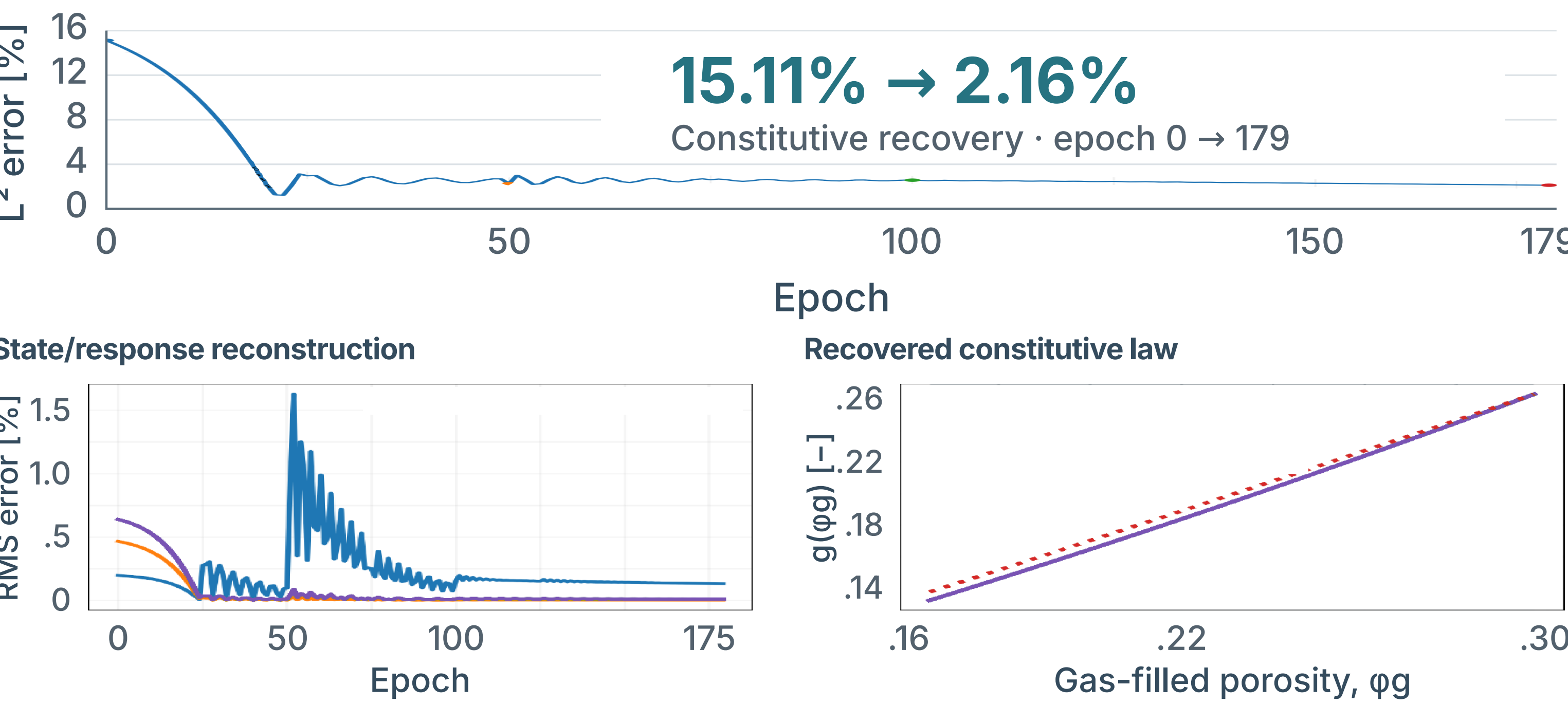
02 Water accumulation grows at high current

Maximum liquid saturation increases nonlinearly at high current density in the synthetic reference benchmark.



03 Recover the missing transport relation

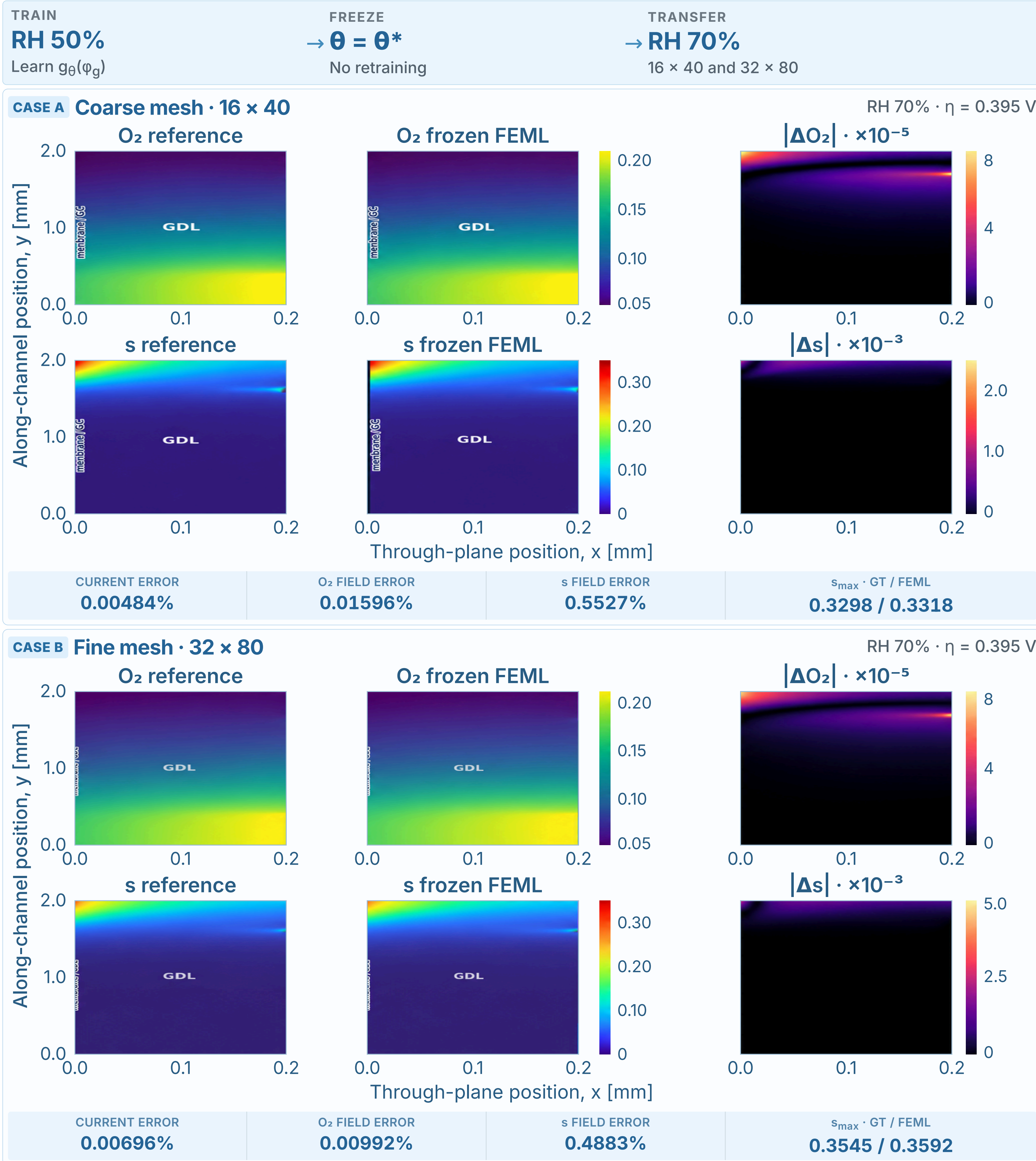
At training RH 50%, the embedded neural closure learns the hidden gas-phase transport relation from synthetic FEM targets.



Field and current reconstruction are evaluated separately from closure recovery. The recovered relation is compared with the synthetic truth over the plotted support. Training reduces the constitutive relative L² error from 15.11% to 2.16%. Insets show reconstruction and closure recovery over the sampled gas-filled-porosity range.

04 Zero-shot transfer with a frozen closure

One learned closure, tested at unseen humidity on two discretizations.



Zero-shot transfer: the same frozen closure reproduces O₂ transport and localized water accumulation on both meshes. Reference-FEML pairs share scales; error maps show pointwise absolute differences. This tests discretization transfer, not mesh convergence. x is stretched for display. Field errors: relative normal L_2 norm. Current error: relative absolute difference. All errors compare frozen FEML with the synthetic reference.

Conclusions

- 01 Physics-preserving learning.** Governing equations remain in FEM; only the uncertain gas-phase closure $g_\theta(\phi_g)$ is learned.
- 02 Constitutive recovery.** 2.16% relative L² error at epoch 179 over the evaluated support.
- 03 Frozen transfer.** Agreement at RH 70% on both 16×40 and 32×80 , without retraining. Saturation-field errors: 0.55% coarse $\cdot$ 0.49% fine. Synthetic verification only. Experimental validation and wider operating-domain tests remain future work.